

CLAUDE CODE

Claude Code 15: The Results Are In: Can LLMs Replicate a PNAS Paper? (Part 2)

69% agreement, \$11, and 2.6 hours later



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FEB 06, 2026



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TL;DR

I used gpt-4o-mini to replicate the text classification from [Card et al.'s PNAS paper 140 years of immigration rhetoric](#). Here's what happened:

- **Total Cost of this Exercise:** \$10.99
- **Total Time:** 1 hour set up (yesterday's video), 2.6 hours (we expected 24), 1 hour analysis today = 4.6 hours.
- **Agreement with original classifier:** 69%
- **Key finding:** The polarization story is robust—both classifiers show Democrats and Republicans diverging sharply since the 1970s

Bottom line: LLMs can substantially replicate trained classifier results at a fraction of the cost. The substantive conclusions survive. If you're doing exploratory text analysis without resources to fine-tune a custom model, this is now a viable path.

The catch: The LLM is more "cautious"—when uncertain, it calls things NEUTRAL instead of picking a side. Direct PRO↔ANTI disagreements are rare (only 4%).

The video is below, and a word of thank you to all paying subscribers and those who have gifted the substack and shared the posts. Thank you! This substack is a labor of love, and I really enjoy sharing what I'm learning about using Claude Code for quantitative social scientific research. So thank you everyone for your support.



Now for the details.

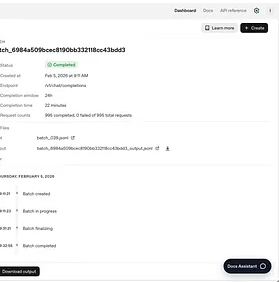
In Part 1 (link below), I described an experiment: could Claude Code help me [replicate the text classification from Card et al.'s PNAS paper on 140 years of immigration rhetoric](#)? And in yesterday's post, Claude Code web crawled to find the replication package, organized my directories, came up with a plan, cracked open the files, submitted json files in batch to OpenAI to have the gpt-4o-mini do a one-shot analysis of 304,995 speeches ready to classify. I used a batch job which is 50% cheaper than

doing it one request at a time, plus it's 305k speeches so you can't realistically do that anyway. But there was a lot of uncertainty as to what would happen.

CLAUDE CODE

Claude Code Part 14: I Asked Claude to Replicate PNAS Paper Using OpenAI's Batch API. Here's What Happened (Part 1)

SCOTT CUNNINGHAM • FEB 5



I've been experimenting with Claude Code for months now, using it for everything from writing lecture slides to debugging R scripts to managing my chaotic academic life. But most of those tasks, if I'm being honest, are things I could do myself. They're just faster with Claude.

[Read full story](#) →

Originally, yesterday, Claude Code told me that it would take 24 hours for the analysis to be completed, but it actually took only 2.6 hours and probably less because I had resubmitted a couple of batches that had failed and only discovered that was the case around 2.5 hours. So I wasn't sure how long it had been done is my point when I resubmitted in those last two batches. Anyway, it's done, the results are in, and I filmed myself analyzing it, as well as creating a [deck](#) (which I go through in the video).



Deck Final
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Here's what we found.

The Timeline: Way Faster Than Expected

First, just so this has its own section and like I said, when I submitted the batch job expected to wait. And that's because OpenAI's documentation says batch jobs can take "up to 24 hours." I thought originally that when Claude Code told me, therefore, it would likely take 24 hours, it was because he did some back of the envelope calculation, but no. he was just relying on the time stamp. I figured I'd check back tomorrow next morning. But when I checked the time stamps, it had taken around 2 1/2 hours to do it.

That's not a typo. **Two and a half hours to classify nearly 300,000 speeches.**

For context: the original paper's authors had to fine-tune a RoBERTa model on 7,626 human annotations, then run inference on their entire corpus. That's weeks of work if you include the annotation time, maybe more depending on how many RAs they probably had to hire to do that annotation.

But that's not what we did. I actually originally thought in fact that we were using 7,626 human annotations in this exercise, but we weren't. Rather we did a zero-shot replication. For eleven dollars.

The Bottom Line: 69% Agreement

Let me give you the headline number first, then we'll unpack what it means.

gpt-4o-mini agreed with the original RoBERTa classifier on 69% of speeches.

Is that good? It depends on your baseline.

Here's what helped me calibrate: the original paper reports that *human annotators* agreed with each other at Krippendorff's $\alpha = 0.48$. That's considered "moderate" agreement in content analysis. Humans themselves often disagreed about whether a speech was pro-immigration, anti-immigration, or neutral.

If even humans can't reliably agree, how much should we expect two different ML systems to agree?

69% isn't perfect. But it's far better than chance (which would be ~33% for a three-class problem). And it's in the ballpark of what you'd expect given the underlying ambiguity of the task.

The Transition Matrix: Where They Agree and Disagree

Here's the full picture:

National Transition Matrix

Original	LLM Classification			N
	Pro	Neutral	Anti	
PRO	63.0%	33.3%	3.7%	102,725
NEUTRAL	8.2%	85.0%	6.7%	136,147
ANTI	4.9%	43.8%	51.3%	86,887

Read: "Of speeches RoBERTa labeled ANTI, LLM classified 51% as ANTI."

Let me walk you through how to read this.

Each row is an original RoBERTa label. Each column is what gpt-4o-mini classified as. The percentages are row percentages—of all speeches RoBERTa labeled as X, what fraction did the LLM label as Y? They must sum to 100% reading left to right because they are treating the original classification as the “sample space” so to speak and then breaking up the share of total new classification into the three different categories, all of which are mutually exclusive.

The diagonal is agreement between old and new classifications. NEUTRAL has the highest agreement (85%). PRO has decent agreement (63%). ANTI has the lowest (51%).

The off-diagonal tells you about systematic disagreements. And here’s the interesting pattern:

When gpt-4o-mini disagrees with RoBERTa, it almost always moves toward NEUTRAL.

- Of speeches RoBERTa called PRO: 63% the LLM also called PRO, but 33% the LLM called NEUTRAL
- Of speeches RoBERTa called ANTI: 51% the LLM also called ANTI, but 44% the LLM called NEUTRAL

The LLM is more *cautious*. When it’s uncertain, it hedges toward the middle category.

The Reassuring Finding: Direct Flips Are Rare

Here’s what worried me going in: what if the two classifiers fundamentally disagree about tone? What if speeches RoBERTa called PRO-IMMIGRATION, the LLM called ANTI?

That would be a serious problem. It would mean the classifiers have incompatible understandings of political language.

The data is reassuring:

- PRO → ANTI: 3.7%
- ANTI → PRO: 4.9%

Direct polarity flips are rare. When the classifiers disagree, they usually disagree about whether something is neutral vs. opinionated—not about which direction the opinion points.

This matters for the substantive findings. If you're using these classifications to track partisan polarization over time, what you really care about is whether a speech is clearly pro, clearly anti, or ambiguous. The two classifiers mostly agree on that structure.

The Figures: Same Story, Different Magnitude

Let me show you Figure 1 from the original paper:

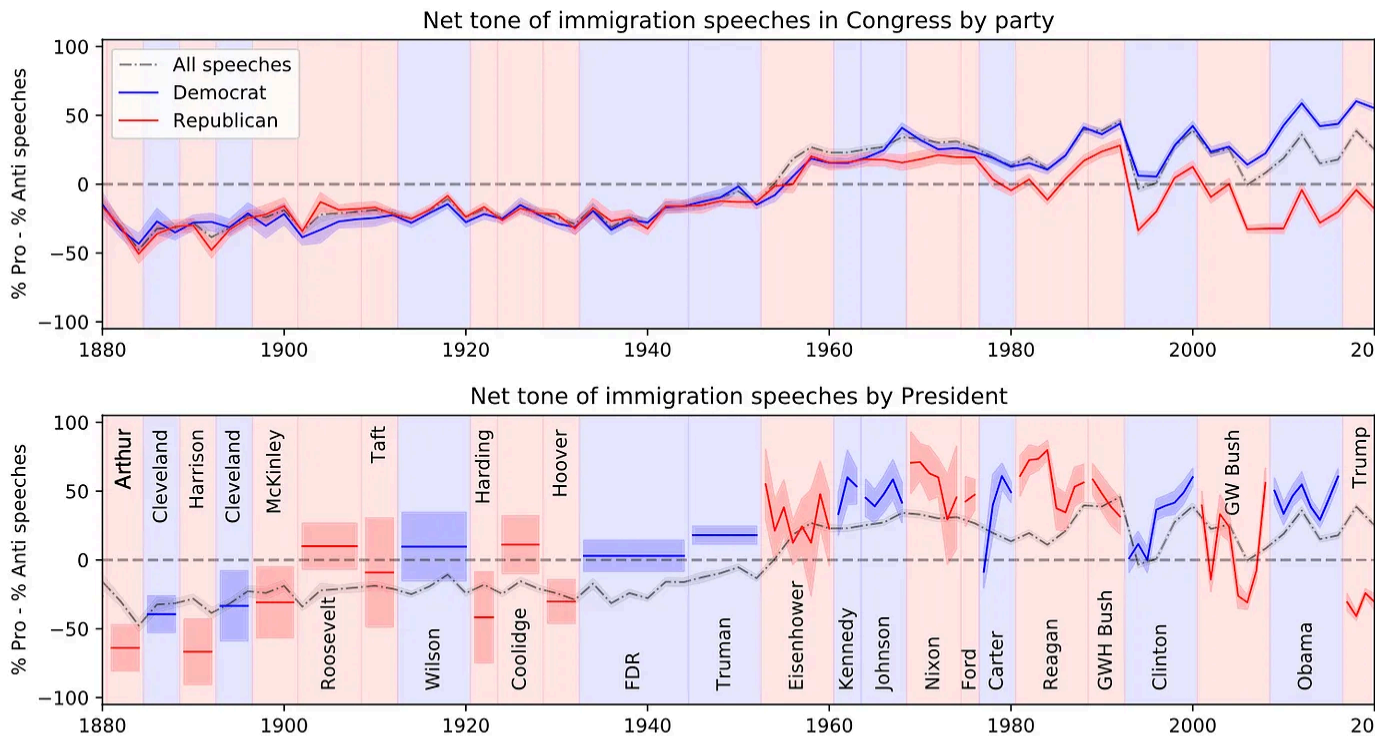
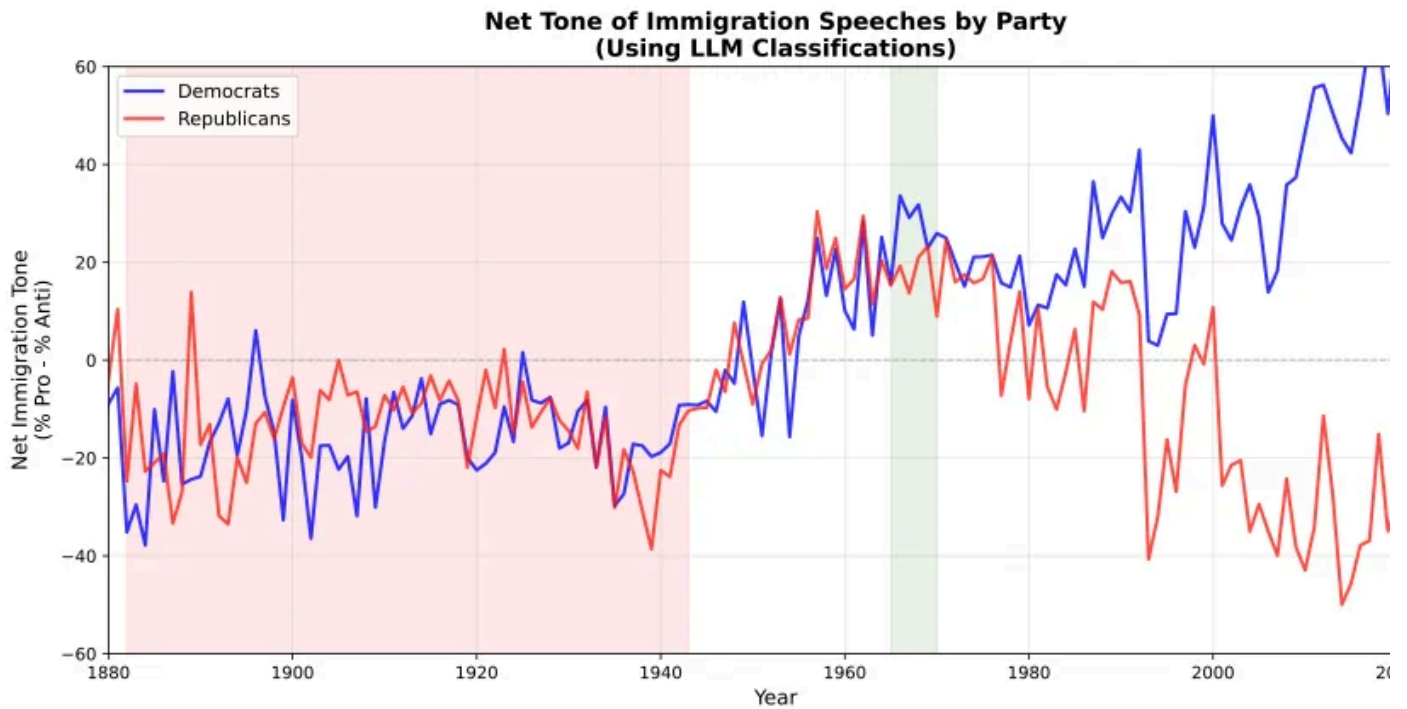


Fig. 1. Evolution of attitudes toward immigration expressed in congressional speeches and presidential communications. Average tone is computed as percentage of proimmigration speeches minus the percentage of antiimmigration speeches, where proimmigration means valuing immigrants and favoring less restricted immigration and vice versa. *Top* and *Bottom* show the overall tone using all congressional speeches about immigration (black dashed line, shaded bands showing plus or minus two SDs based on the estimated proportions and number of speeches). *Top* also shows separate plots for speeches by Democrat and Republicans in Congress. (Due to limitations of the data, about 15% of speeches do not have a named speaker or party affiliation.) *Bottom* shows corresponding estimates for each president, showing the overall average for a president's tenure when there are insufficient data to show annual variation. Note that most modern presidents have been more favorable toward immigration than the average member of Congress. By contrast, Donald Trump appears to be the most antiimmigration president in nearly a century. Similarly, congressional Republicans over the past decade have framed immigration approximately as negatively as the average member of Congress did a century earlier.

This is the key finding: Democrats and Republicans have polarized sharply since the 1970s. Democrats use increasingly positive language about immigrants. Republicans use language as negative as the 1920s quota era.

Now here's the same figure using our LLM classifications:



The polarization story is robust.

Both classifiers show the same basic pattern:

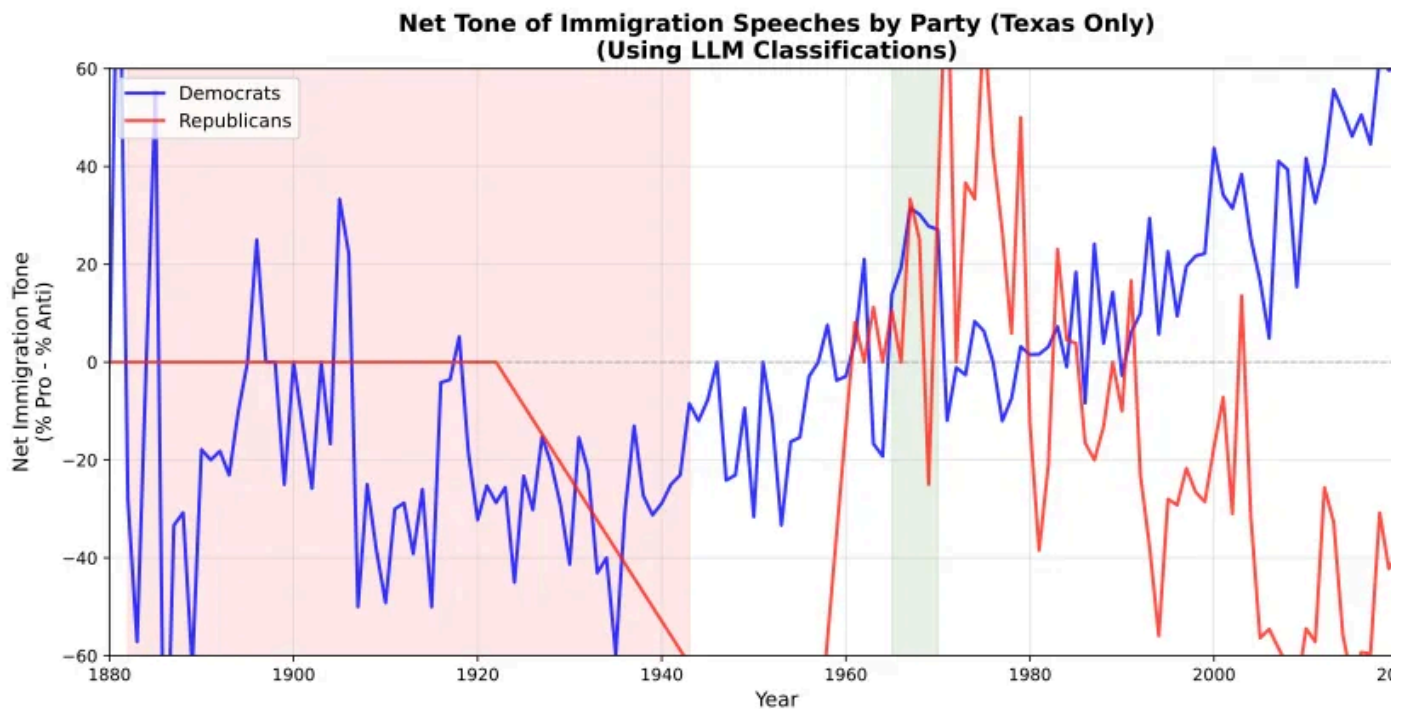
- Partisan divergence starting in the 1970s
- Democrats trending more positive
- Republicans trending more negative
- The gap widening over time

The magnitudes differ—the LLM version shows more volatility, especially in the early period. But the *shape* of the story is the same.

This is the most important finding for anyone wondering whether LLM-based classification can replicate traditional NLP results. The qualitative conclusions hold up. A researcher using gpt-4o-mini would reach the same substantive interpretation as the original authors.

Texas: A Border State Perspective

I was curious whether the patterns looked different for my home state, Texas, so I ran the same analysis restricted to Texas congressional speeches:

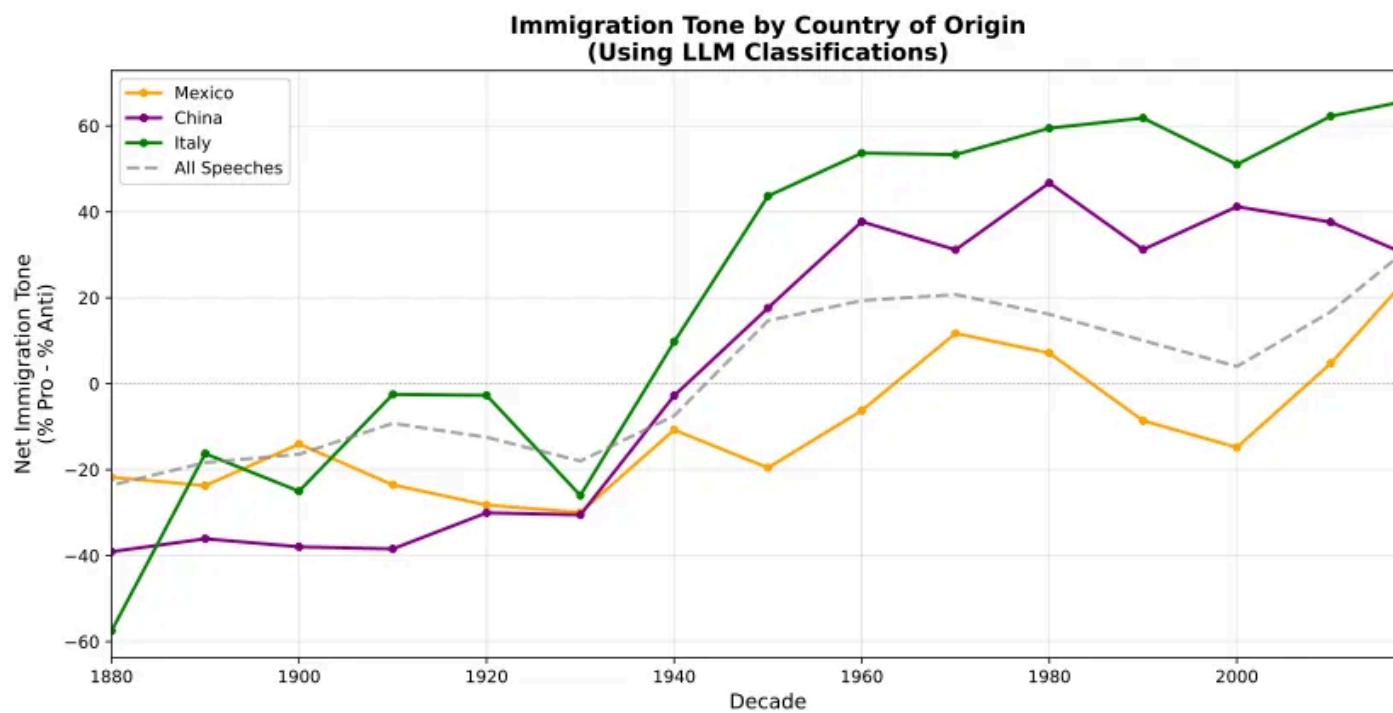


It's noisier, and I think there must not be many Republican Texas congressional speeches in the data before the second half of the 20th century. And when there are speeches by Texans, they are Democrats and interestingly, they are negative toward immigration. But that changes after the war and Democrats in Texas become increasingly more positive. There's a big pro-immigration Texas spike in the mid-1970s, but the same polarization gap happens with Texas as happens nationally starting around 1980.

So, the polarization story holds, though with more noise (smaller sample). And Texas being more negative towards immigration appears to be a bit of a complex historical story that I would love to dig into more but probably won't.

Country of Origin: The Patterns Persist

The original paper also analyzed how tone varied by the country of origin being discussed—Mexico, China, Italy, etc. We replicated that:



Now compare to the original:

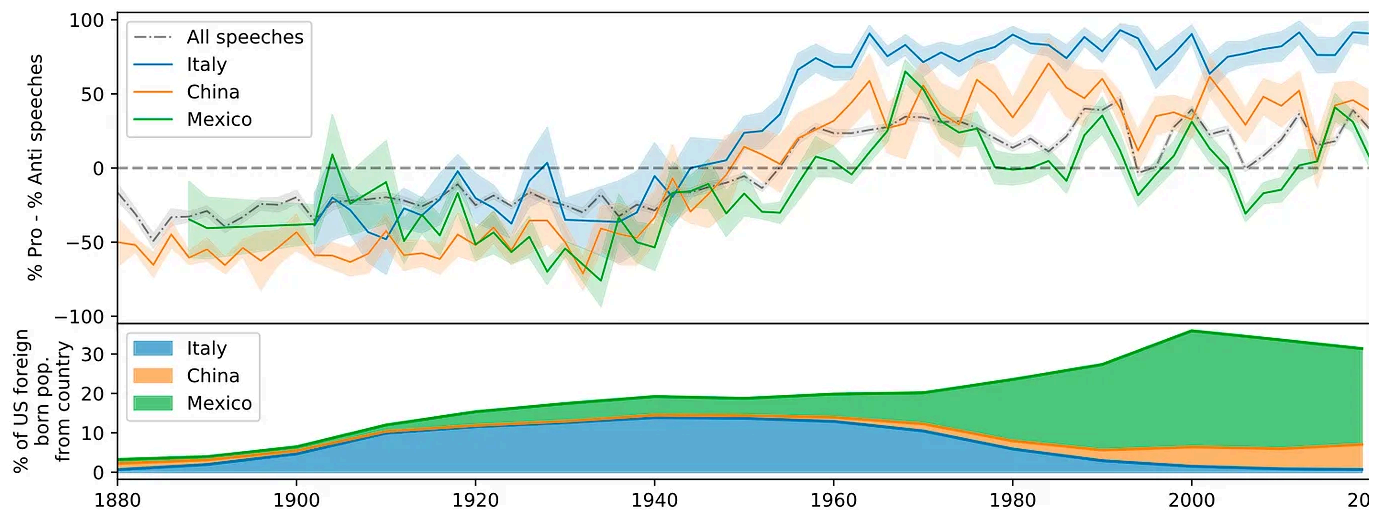


Fig. 2. Average tone of immigration speeches when considering only those speeches that mention the country or nationality for each of the three most frequently mentioned nationalities (*Top*) and the percent of the US foreign-born population from each of these countries over time (*Bottom*). Despite midcentury increase in proimmigration attitudes applying to all groups, a gap in tone by group persists to the present day, with Mexican immigrants being consistently framed more negatively than others and Italian immigrants being framed especially positively. These trends are mirrored in broader regional patterns for Europe, Asia, and Latin American and the Caribbean (*SI Appendix*).

Notice how it isn't merely that the overall trends match between the original Rober classification. Rather, it's even the ordering that's the same. Notice how in the original it was Italy>China>Mexico, which is the same ordering as the LLM classification. Other findings:

- **China:** Most negative in the early period (Chinese Exclusion Era), improving after the 1960s
- **Mexico:** Relatively stable historically, with a dip in the modern era (border politics)
- **Italy:** Most positive overall, especially after WWII ("model minority" framing)

And the LLM captured the country-specific patterns without any fine-tuning. It understood that speeches mentioning "Chinese" in 1890 had different connotations than speeches mentioning "Chinese" in 1990.

Why Would They Disagree?

Let me be clear about something: this was not an apples-to-apples comparison.

RoBERTa (the original):

- Fine-tuned on 7,626 human-labeled examples
- Trained specifically for this task
- Optimized to match annotator judgments
- ~65% accuracy on held-out data

gpt-4o-mini (our replication):

- Zero-shot (no task-specific training)
- General language model
- Inferring the task from a prompt
- No exposure to the original annotations

The RoBERTa model was *taught* what pro-immigration and anti-immigration mean in the context of this specific corpus. gpt-4o-mini had to figure it out from first principles.

Given that asymmetry, 69% agreement is actually impressive. The LLM brought genuine language understanding to the task, not just pattern matching on the specific features the RoBERTa model learned.

Best I can tell, what is going on here is that when they disagree, it's at these "marginal speeches" that were right on the edge between anti and neutral, or pro and neutral. And when you reclassify them, you're mostly pushing the marginal speeches into the neutral territory. Which I think probably means they are more or less random in the data, and as such the signal coming through from the pro and anti in the original classification is pretty strong leaving the overall patterns intact.

What We Learned: The Research Question

1. LLMs can substantially replicate trained classifier results.

69% agreement isn't perfect, but it's good enough that the substantive findings survive. If you're doing exploratory text analysis and don't have resources to fine-tune a custom model, gpt-4o-mini is a viable option.

2. The polarization finding from Card et al. is robust.

Both classifiers tell the same story about partisan divergence. This increases my confidence in the original paper's conclusions. The finding isn't an artifact of their specific model.

3. LLMs are more conservative.

gpt-4o-mini preferred NEUTRAL when uncertain. This is probably good behavior—it's better to say "I don't know" than to confidently guess wrong. But it means LLM based classification may underestimate the proportion of clearly opinionated text.

4. Direct polarity flips are rare.

The classifiers rarely called the same speech both PRO and ANTI. When they disagreed, it was usually about whether something was neutral vs. opinionated. This is reassuring for anyone using these tools for sentiment analysis.

5. LLM was cheaper

And perhaps the most important thing — the cost. Something like 4 hours, start to finish, and only \$11. The hard part was therefore probably in getting the data — but I bet you that too is a quest we could undertake at considerable speed given it's entirely possible these are available to us such that Claude Code would've crawled and scraped the speeches, or they are stored in some warehouse somewhere. Though maybe this team themselves transcribed all 305k of them; I'll need to check more closely to see

think still that the real bottleneck is going to be on the margins of getting the data any DUAs that don't allow you to use an AI agent for analysis, but I still believe we see creative ways where researchers accommodate their DUAs to the letter but we use AI agents to speed things up. For instance, just the creating of the audited code ante before you enter some closed off location is itself alone going to be possible.

What We Learned: The Process

Beyond the substantive findings, this experiment taught me things about using LLM for research:

- **The Batch API is underrated.** Most researchers I know use ChatGPT interactively. *Maybe*, they call the API one request at a time. But that's a big *ma* Using the Batch API is cheaper (50% off), handles large jobs gracefully, and finished way faster than promised. If you're doing any kind of text analysis at scale, learn the Batch API. *But you can also have Claude Code do it for you and explain it to you in the way your brain learned along the way.*
- **Referee 2 was essential.** Having a separate Claude instance review the code caught real bugs: edge cases in label normalization, missing metrics (Cohen's Kappa), concerns about prompt design. The code that ran was better than the c I would have submitted without review.
- **Not sticking to my planned workflow.** But at the same time, yesterday I inadvertently used referee 2 from within the same context window which as I s violates the entire point of using referee 2. We don't want students grading the own exams. And we don't want audits to be done by the very Claude Code manifestation that wrote the code. So somehow there is a mental thing about tl referee 2 workflow that is still not clicking for me, and I'm not sure what it is. I don't know if it's because I'm filming myself working, or what, but the point is, you won't get an error when you audit your code using referee2 from the incorr

Claude Code context window. Which means that I can audit my code incorrect and never know. I only know because a reader/viewer caught it and told me, which has given me pause about how workflows will work now.

- **\$11 is not a lot of money.** The cost barrier for this kind of research is now trivial. The barrier is knowing how to set it up—which is why I wrote this.
- **2.6 hours is not a lot of time.** I expected to wait overnight. I barely finished dinner. If you're iterating on prompt designs or testing different models, you can run multiple experiments in a day.

The Cost and Time Summary

Let me put the practical numbers in one place:

What This Cost

Item	Value
Speeches classified	296,995
Total tokens	~142 million
Time to complete	2.6 hours
Total cost	\$10.99

Less than a Netflix subscription to replicate a PNAS paper.

That catchphrase in the caption — Less than a Netflix subscription to replicate a PNAS paper — was Claude Code’s own astonishment at this whole thing!

What This Means

Three years ago, if you wanted to classify 300,000 text documents, you had two options: hire human annotators (expensive, slow) or train a custom NLP model (requires expertise, compute, labeled data).

Now there’s a third option: send your documents to a general-purpose LLM and get results in a few hours for pocket change. **And use Claude Code (or Codex or something like them both) to help you do it so you don’t mess it up.**

The results won’t be identical to a fine-tuned model, but that’s the point. It’s a different NLP classification. This exercise wasn’t about trying to speak into that, though. I really didn’t know what I was going to find. I just wanted to show that Claude Code pulled all this together from the replication package it found itself online, set it up and submitted the batches to openai. All for only \$11 and around 4 hours time start to finish, including making the cool decks to teach me it. For many research questions, that’s good enough for just gaining quick and lasting insights.

This changes who can do computational text analysis. You don’t need a machine learning background. You don’t need GPU clusters. You don’t need a budget for annotation.

You need a research question, some text data, and eleven dollars.

Resources

Original paper:

Card, D., Chang, S., Becker, C., Mendelsohn, J., Voigt, R., Boustan, L., Abramitzky, I. & Jurafsky, D. (2022). Computational analysis of 140 years of US political speeches reveals more positive but increasingly polarized framing of immigration. *PNAS*, 119(31), e2120510119.

Replication data:

github.com/dallascard/us-immigration-speeches

Our replication:

- Model: gpt-4o-mini via OpenAI Batch API
- Cost: \$10.99
- Batch Processing Time: 2.6 hours
- Agreement: 69%

This is Part 2 of a two-part series on using LLMs for research replication. Part 1 covered the setup; this part covered the results. If you're interested in trying this yourself, the key tools are Claude Code for orchestration, the OpenAI Batch API for scale, and a healthy dose of methodological skepticism.



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Dr Sam Illingworth  Slow AI 3h

 Liked by [scott cunningham](#)

Scott. This is an absolute game-changer for empirical research. Do you think that there's a way we could actually use this in the peer review process to independently verify results in a way that's possible never been possible before?

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2 replies by [scott cunningham](#) and others

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